

Modernizing Biopharmaceutical Manufacturing: From Batch to Continuous Production

The importance and value of continuous bioprocessing, both up and downstream. Economic and sustainability advantages and due to the modular nature of continuous bioprocesses means that the industry is able to adapt more rapidly to changing market demands. Continuous biopharmaceutical manufacturing in the context of other industries that have already successfully adopted continuous processing. Factor other than scientific ones, are the barriers to change from batch to continuous production. An excellent example of the manufacturing strategies of the steel industry in the 20th century, when this industrial sector incrementally switched from batch to continuous operations. Biopharmaceutical industry has reached a stage that requires a change in the production paradigm. For a certain class of biopharmaceutical products upstream continuous manufacturing has always been applied: for example, unstable proteins that rapidly degrade in the culture broth. In order to obtain a high quality product, the residence time in the reactor must be minimized. This can only be achieved with continuous cultivation and preferably with perfusion reactors.

Furthermore, this is a universal production platform that can be extended to other classes of products, such as antibodies, which are relatively stable molecules. Continuous manufacturing is as productive and with a much smaller footprint of the manufacturing plant, avoiding multiple non-value added unit operations. In essence, the investment for a continuous plant is much smaller compared to a batch-operated one.

Optimization of present conventional batch biomanufacturing

The traditional batch manufacturing is gone through every possible improvement step to optimize the process, decrease complexity, reduce Capital Expenditure (CAPEX) and Operational Expense (OPEX) that could improve and streamline the biomanufacturing of drug substance and product delivery to patients. The industry implemented many of the available technologies and operational management tools to achieve this task;

- Increase "Overall Asset Effectiveness" (OAE)
- Reduce "Throughput Time" (TPT)
- Improve "Quality" (Q) by using seamlessly integrated and



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well characterized processes

Implementing Lean manufacturing and other optimization programs to improve traditional batch biomanufacturing resulted in some improvements but still struggling with;

- Operation
 - Major "efficiency gains" have been implemented
 - Additional "quantum leap efficiency gains" are unlikely
 - "Manufacture and ship" across continent
- Compliance
 - Many "manual" checks
 - Deviations and investigations

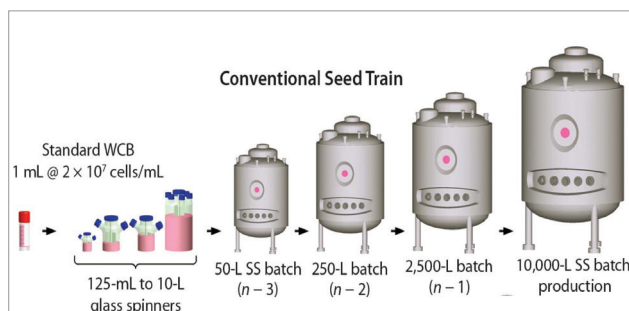


Figure 1: Typical conventional production scale fed-batch system, upstream (the production scale reactor could be up to 30 KL)

- o Difficult “root cause” analysis
- Quality
- o Reliance on “in-process” and “end-product” testing

Upstream Fed-Batch system: an overall improvement resulted due to achieving higher manufacturing titer (viable cell density) figure 5, improved controls (PAT, QbD), flexible equipment and system designs (utilizing Single use technology where applicable) that progressed the biomanufacturing from an overall complex line shown in figure 1, to an improved simpler operation shown in figure 2.

For upstream perfusion system an overall improvement resulted due to better titer, higher volumetric flow-rate (volumetric productivity) figure 6, flexible equipment and better system designs from the overall complex operation shown in figure 3, to the simpler operation shown in figure 4.

Continuous biomanufacturing

To distinguish between Batch manufacturing and continuous manufacturing: In batch manufacturing, all materials are charged before the start of processing and discharge at the end of processing. In continuous manufacturing, materials are simultaneously charged and discharged from the process.

To achieve upstream continuous biomanufacturing, it requires a paradigm shift thinking and an implementation of systems that is;

- Modular
- Flexible
- Portable
- Compact
- That requires less real estate
- Cost effective
- Optimized
- And friendly on change-over

This requires a design and implementation of new technologies to attain the goal as illustrated in the following example, in table 1;

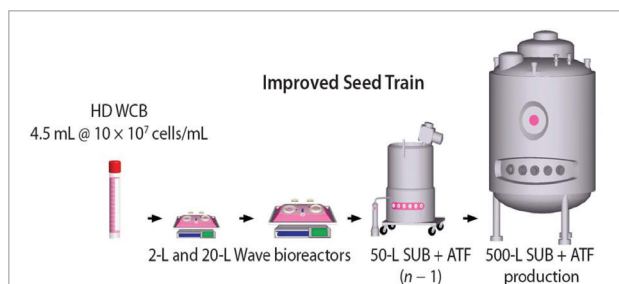


Figure 2: Typical Production scale improved fed batch system, upstream (the production scale reactor could be up to 2 KL)

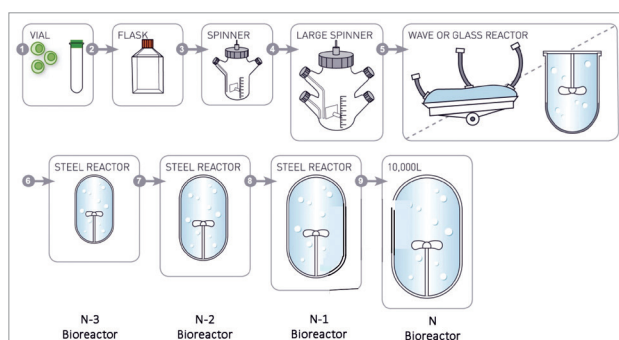


Figure 3: Typical conventional production scale perfusion system, upstream

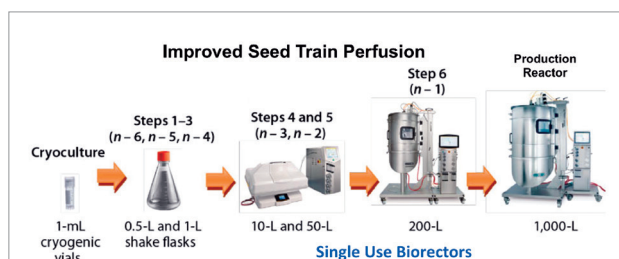


Figure 4: Typical improved Production scale perfusion system, upstream

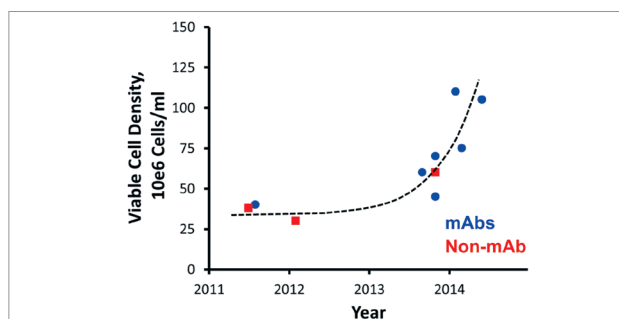


Figure 5: Viable Cell Density, Rapid progress in upstream production capability, significant upward potential²¹

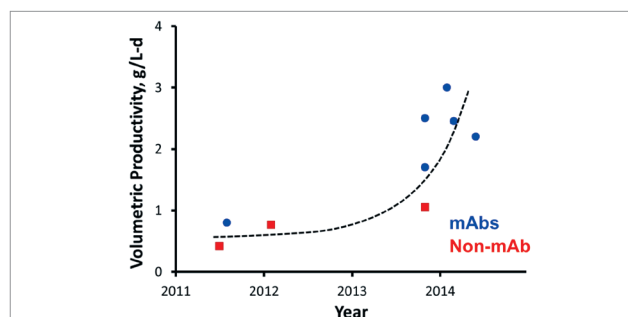


Figure 6: Volumetric Productivity, rapid progress in upstream production capability, significant upward potential²¹

By implementing what outlined in table 1 and clinical experimental data (proof of concept reported by few firms that conducted the studies) the upstream system would result in the simple system as illustrated in figure 7.

Process Design

Measurement and control for enabling continuous processing adoption in the biotech and biopharmaceutical industries – the shift from batch to continuous production methods is transforming the future of biopharmaceutical manufacturing. The potential economic gains from increasing capacity utilization and reducing the length of process development, product release times and capital costs are driving the paradigm shift. Spurred by potential improvements to quality, patient safety and the time required for breakthrough medicines to reach patients.

The U.S. Food and Drug Administration (FDA) have advocated a move to continuous manufacturing (CM). However, with new methods come new challenges — particularly for process design, measurement, material traceability and control. Manufacturers, suppliers and research institutions are collaborating to solve these challenges on projects across the globe.

The status of continuous bioprocessing in the

biopharmaceutical industry is based in part on research on how manufacturers can overcome the challenges related to quality, compliance, material traceability and process design and control.

Continuous Biomanufacturing - Upstream

A process platform for a fully integrated continuous fermentation, clarifications, harvest and capture of the desired protein consists of a stirred tank reactor with a cell retention device, figure 7, the reactor is continuously harvested and the cell-free effluent captured by a continuous chromatographic process. In such a process, the culture supernatant is loaded to full saturation on a chromatography column and after breakthrough; the column effluent is loaded on a second column. After full saturation the column is disconnected from the second column and product is recovered, whereas the second one is loaded to full saturation. After breakthrough of the second column the effluent is loaded onto a third column. The second column is disconnected again and product is recovered. This process is also known as countercurrent chromatography and has been implemented at pilot scale. The system is more complex than a typical batch chromatography process, but fully automated and with higher productivity. The product quality could be maintained at high levels during long-term steady state operation of the integrated continuous system. In light of the current transition of biopharmaceutical industry towards Quality-by-Design and real-time product release, continuous manufacturing technology is a big step in getting closer to this ambitious goal.

Continuous Biomanufacturing - Downstream

Downstream Processing focuses on yield and productivity as well as on purity and process capacities. An increase in separation efficiency of single unit operations is achieved by expansion of existing facilities and by optimization of existing and alternative processes². These include the establishment of platform technologies, high-through-put methods with approaches based on QbD and DoE-based experimental optimizations^{2,10,12}, additionally, an integration of modeling and simulation of unit operations as well as the use of mini-plant facilities.

Traditionally, monoclonal antibodies were purified by a sequence of different chromatographic and membrane-based operations^{2,4,5,6,7,13,14}. A virus-inactivating operation, a filtration-based virus-reducing step and a final

Action	Key Technology	Benefit
- Steady State - High Cell Density (Titer) - High Productivity (Volumetric Flow Rate)	- High Speed Production Rate - Low Perfusion Rate	- Reduced Bioreactor Size - SUBs Now Feasible - Standardized Size - Universal – mAb/Non-mAb
Continuous Flow from Bioreactor to Capture	Simultaneous Cell Separation and Clarification	- Removes: - Hold Steps, Hold Tanks - Clarification Operations - Simplified Process
Continuous Capture	Periodic Countercurrent (PCC) Chromatography	Reduced Column Size and Buffer Usage

Table 1: Upstream Continuous Manufacturing: Action, Technology and Benefit

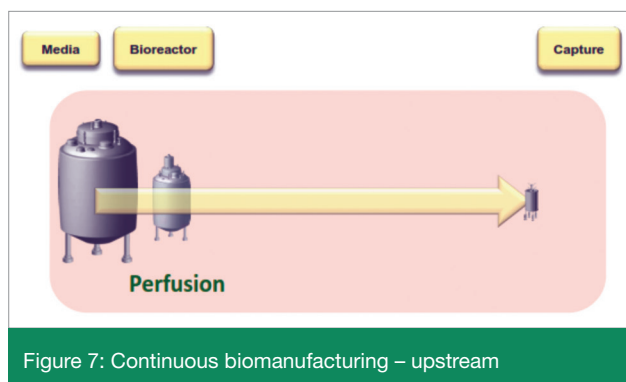


Figure 7: Continuous biomanufacturing – upstream

diafiltration have to be included^{1,4,7,8,13,14}.

After cell harvesting by centrifugation and or filtration, a chromatographic separation unit is used to isolate antibodies from fermentation broth^{2,5,7,8,14,15}. Protein A chromatography is one of the most important unit operations for antibody capturing^{7,15,16}. It distinguishes itself by high selectivity towards IgG-type antibodies, high flow rate and capacity. The dynamic binding capacity ranges from 15–100 g mAb/L resin depending on antibody, flow rate and adsorbent^{7,17,18}. The degree of purity is consistently higher than 95%^{3,15,16,19}. Process-related impurities like DNA (Deoxyribonucleic Acid), HCP (Host Cell Proteins), media components and virus particles are removed^{7,15,16}. One of the major advances in recent years of process development consists in a better integration of chromatography to the overall manufacturing process. Elution conditions of the initial Protein A capture step are adjusted to the following unit operation in order to enter a subsequent virus inactivation step or an ion exchange chromatography⁷. This eliminates any need for buffer exchange between these unit operations and it is one example for a successful integration of single separation operations during the last decades.

Problems exist in form of Protein A leaching and non-specific binding of impurities like HCP and DNA. Leached Protein A reduces the binding capacity of Protein A chromatography and needs to be removed in subsequent purification steps^{7,16,20}. The amount of bound impurities depends on the adsorbent, composition of cell culture harvest, column loading and washing conditions^{7,16}. Tarrant *et al.* (2012)¹⁶ and Shukla *et al.* (2008)¹⁵ published studies on HCP interacting with different Protein A matrices and the product.

Continuous manufacturing is currently being seriously considered in the biopharmaceutical industry as the possible new paradigm for producing therapeutic proteins, due to production cost and product quality related benefits.

An example⁴ monoclonal antibody producing CHO cell line was cultured in perfusion mode and connected to a continuous affinity capture step. The reliable and stable integration of the two systems was enabled by suitable control loops, regulating the continuous volumetric flow and adapting the operating conditions of the capture process. For the latter, an at-line HPLC measurement of the harvest concentration subsequent

to the bioreactor was combined with a mechanistic model of the capture chromatographic unit. Thereby, optimal buffer consumption and productivity throughout the process was realized while always maintaining a yield above the target value of 99%. Stable operation was achieved at three consecutive viable cell density set points (20, 60, and 40×10^6 cells/mL), together with consistent product quality in terms of aggregates, fragments, charge isoforms, and N-linked glycosylation. In addition, different values for these product quality attributes such as N-linked glycosylation, charge variants, and aggregate content were measured at the different steady states. The amount of released DNA and HCP was significantly reduced by the capture step for all considered upstream operating conditions.

In the current environment of diverse product pipelines, rapidly fluctuating market demands and growing competition from biosimilars, biotechnology companies are increasingly driven to develop innovative solutions for highly flexible and cost-effective manufacturing. To address these challenging demands, integrated continuous processing, comprised of high-density perfusion cell culture and a directly coupled continuous capture step, can be used as a universal biomanufacturing platform.

Another example⁵, reports the first successful demonstration of the integration of a perfusion bioreactor and a four-column periodic counter-current chromatography (PCC) system for the continuous capture of candidate protein therapeutics. Two examples are presented: (1) a monoclonal antibody (model of a stable protein) and (2) a recombinant human enzyme (model of a highly complex, less stable protein). In both cases, high-density perfusion CHO cell cultures were operated at a quasi-steady state of $50\text{--}60 \times 10^6$ cells/mL for more than 60 days, achieving volumetric productivities much higher than current perfusion or fed-batch processes. The directly integrated and automated PCC system ran uninterrupted for 30 days without indications of time-based performance decline. The product quality observed for the continuous capture process was comparable to that for a batch-column operation. Furthermore, the integration of perfusion cell culture and PCC led to a dramatic decrease in the equipment footprint and elimination of several non-value-added unit operations, such as clarification and intermediate hold steps. These findings demonstrate the potential of integrated continuous bioprocessing as a universal platform for the manufacture of various kinds of therapeutic proteins.

Conclusion

Traditional batch biomanufacturing is restrictive

Inherent obstacles in batch biomanufacturing:

- Capital Intensive
- Consumes Large Real Estate
- Requires enormous time to build the facility

- Requires large Staff to operate
- The demand for capacity is going up
- And there is a need for local manufacturing

Disadvantages of batch processing

- Defined batch size (output quantity driven by batch size)
- Multiple, sequential process steps, end to end
- Many interruptions between/during process steps
- Long waiting times between single process steps
- Numerous transport steps between process steps
- Lengthy throughput times from start to finish
- High levels of raw material and intermediate inventories required
- Extensive validation and scale-up activities needed
- Physical and organizational separation in operations and development
- Quality measured by in process sampling/control and end product testing

Integrated continuous biomanufacturing platform in summary

Integrated continuous biomanufacturing, figures 8, 9, and 10; is:

- Universal, Standard platform (various proteins)
- Steady State (metabolism)
- Closed system (minimized microbial issues)
- No scale-up, same scale for pilot & Manufacturing
- Due to scale, compatible with single-use/disposable technology
- Minimized hold time
- Continuous flow
- High volumetric productivity
- Integrated, modular, simplified operation
- Flexible capacity, increase and or decrease

Advantages of Continuous Manufacturing

- Integrated bioprocessing with fewer steps
 - no manual handling
 - increased safety
 - shorter processing times
 - increased efficiency
- Smaller equipment and facility footprint
 - more flexible operation
 - reduced inventory
 - lower costs
 - less work-in-progress material
 - smaller ecological footprint (sustainability)
- On-line monitoring and control for increased product quality assurance in real-time
 - amenable to Real-Time Release Testing approaches
 - consistent quality

CAPEX and OPEX advantages of Continuous Manufacturing;

- Integration of Compliance/Quality within the Process

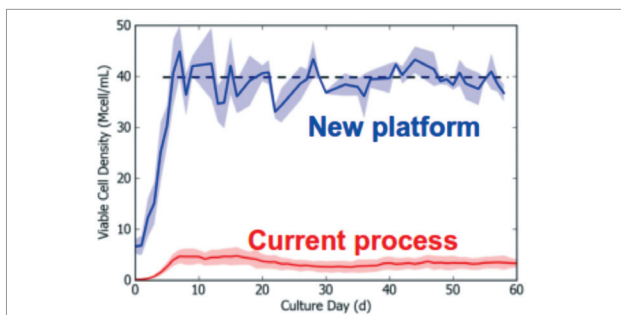


Figure 8: Cell density, current process vs new platform²¹

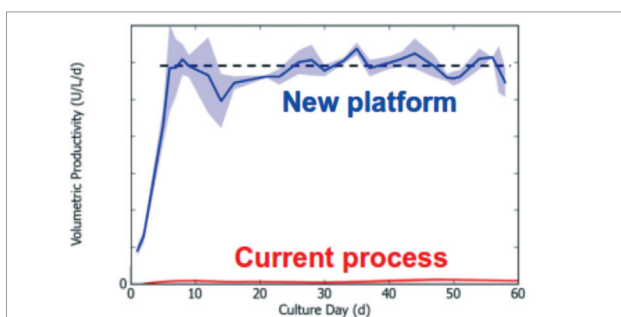


Figure 9: Volumetric production rate, current process vs new platform²¹

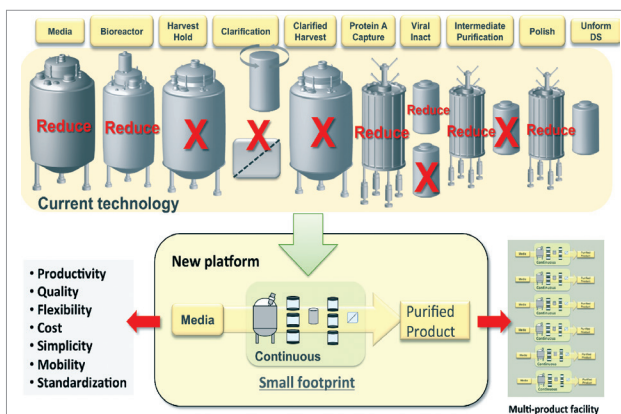


Figure 10: going from traditional batch biomanufacturing to integrated continuous bioprocessing²¹

- Reduction of asset footprint (40-90%)
- Reduction in CapEx (25-60%)
- Reduction of OpEx (25-60%)
- Reduction of raw material and intermediate inventories
- Flexibility in supply size
- Reduction in overall development times for drug substances and drug products and Improving time to market
- Assuring availability of high quality, safe, and efficacious drug products to patients

What we are doing brings technology and innovation to the heart of the manufacturing forefront.

Continuous manufacturing and integrated continuous manufacturing (ICM) have been successfully implemented at certain drug product manufacturers with the assistance of educational institutions and equipment manufacturers / suppliers. This cooperation was reached because of having the same goal and common interest in better and streamlined manufacturing. These processes, under R&D for the last several years, have been under study, and discovery is now a reality. There are some elements which are pretty obviously going to be incredibly important to the biotech industry, and some, in parts, are company secret.

With such compact manufacturing units, biotech companies could make more types of drugs, or they could quickly scale up production of blockbusters by adding units as needed. These new technologies would let firms be much more diverse and adaptive, moving in and out of different products more quickly. More flexible manufacturing could help prevent damaging drug shortages. The FDA reported 251 drugs in short supply in 2011 for medicines that are injected, as most biotech products are. About 20 percent of the time, the problem was that companies' manufacturing capacity fell short.

The biopharmaceutical industry has typically paid little attention to innovations in manufacturing, but because certain firms have kept manufacturing capacity near to their labs in Massachusetts and California, they're now well-positioned to implement new ideas. They have the right scientists, the right engineers and the right suppliers in the U.S. We are hopeful that the U.S. will continue to be a leader in this new technology. What we are doing brings technology and innovation to the heart of the manufacturing forefront. It's not clear how long it will take until patients receive drugs made through the new processes. At least we think that the main technical barriers have been overcome, so it's now a business decision — this is being pushed very hard by a few manufacturers.

References

- Jain, E.; Kumar, A. Upstream processes in antibody production: Evaluation of critical parameters. *Biotechnol. Adv.* 2008, 26, 46–72. [Google Scholar] [CrossRef]
- Shukla, A.A.; Thömmes, J. Recent advances in large-scale production of monoclonal antibodies and related proteins *Trends Biotechnol.* 2010, 28, 253–261. [Google Scholar] [CrossRef]
- Gagnon, P. Technology trends in antibody purification. *J. Chromatogr. A* 2012, 1221, 57–70. [Google Scholar] [CrossRef]
- Chon, J.H.; Zarbis-Papastoitsis, G. Advances in the production and downstream processing of antibodies. *New Biotechnol.* 2011, 28, 458–463. [Google Scholar] [CrossRef]
- Kelley, B. Industrialization of mAb production technology: The bioprocessing industry at a crossroads. *MAbs* 2009, 1, 443–452. [Google Scholar] [CrossRef]
- Low, D.; O'Leary, R.; Pujar, N.S. Future of antibody purification. *J. Chromatogr. B* 2007, 848, 48–63. [Google Scholar] [CrossRef]
- Liu, H.F.; Ma, J.; Winter, C.; Bayer, R. Recovery and purification process development for monoclonal antibody production. *MAbs* 2010, 2, 480–499. [Google Scholar] [CrossRef]
- Birch, J.R.; Racher, A.J. Antibody production. *Adv. Drug Deliver. Rev.* 2006, 58, 671–685. [Google Scholar] [CrossRef]
- Shukla, A.A.; Hubbard, B.; Tressel, T.; Guhan, S.; Low, D. Downstream processing of monoclonal antibodies—Application of platform approaches. *J. Chromatogr. B* 2007, 848, 28–39. [Google Scholar] [CrossRef]
- Bhambure, R.; Kumar, K.; Rathore, A.S. High-throughput process development for biopharmaceutical drug substances. *Trends Biotechnol.* 2011, 29, 127–135. [Google Scholar] [CrossRef]
- Heckathorn, R.; Adams, D.; Hunter, J.; Frieden, E. Increasing Upstream Process Development Efficiency by Implementing Platform Glutamine Synthetase Cell Culture Processes. In *Cells and Culture*; Noll, T., Ed.; Springer Netherlands: Heidelberg, Germany, 2010; pp. 245–251. [Google Scholar]
- Del Val, I.J.; Kontoravdi, C.; Nagy, J.M. Towards the implementation of quality by design to the production of therapeutic monoclonal antibodies with desired glycosylation patterns. *Biotechnol. Prog.* 2010, 26, 1505–1527. [Google Scholar] [CrossRef]
- Gottschalk, U. *Process Scale Purification of Antibodies: Downstream Processing of Monoclonal Antibodies: Current Practices and Future Opportunities*, 1st ed.; John Wiley & Sons Inc.: Hoboken, NJ, USA, 2009. [Google Scholar]
- Sommerfeld, S.; Strube, J. Challenges in biotechnology production—Generic processes and process optimization for monoclonal antibodies. *Chem. Eng. Process.* 2005, 44, 1123–1137. [Google Scholar] [CrossRef]
- Shukla, A.A.; Hinckley, P. Host cell protein clearance during protein A chromatography: Development of an improved column wash step. *Biotechnol. Prog.* 2008, 24, 1115–1121. [Google Scholar] [CrossRef]
- Tarrant, R.D.R.; Velez-Suberbie, M.L.; Tait, A.S.; Smales, C.M.; Bracewell, D.G. Host cell protein adsorption characteristics during protein A chromatography. *Biotechnol. Prog.* 2012, 28, 1037–1044. [Google Scholar] [CrossRef]
- Royce, J. High-capacity protein A chromatography medium for mAb capture from high-titer feeds. *BioProcess Int.* 2014, 12, 40–41. [Google Scholar]
- Lain, B.; Cacciottolo, M.A.; Zarbis-Papastoitsis, G. Development of a high-capacity Mab capture step based on cation-exchange chromatography. *BioProcess Int.* 2009, 26–34. [Google Scholar]
- Lain, B. Protein A: The life of disruptive technology. *BioProcess Int.* 2013, 11, 29–38. [Google Scholar]
- Ghose, S.; Hubbard, B.; Cramer, S.M. Evaluation and comparison of alternatives to Protein A chromatography: Mimetic and hydrophobic charge induction chromatographic stationary phases. *J. Chromatogr. A* 2006, 1122, 144–152. [Google Scholar] [CrossRef]
- Konstantin Konstantinov, VP, Late Stage Process Development, BioRealization, Sanofi, (noted figures, via e-mail).